Macro-Informed Sector Rotation

A Framework for Regime-Aware Tactical Asset Allocation

I. Problem Selection & Framing

The starting point for this project was not a model, but a question: *'Why do simple momentum strategies fail catastrophically at certain times?'* Examining historical data, it becomes clear that static allocation rules break down during regime shifts—periods where the underlying data-generating process changes structurally.

Instead of treating this as a pure return-forecasting problem, I reframed it as a **regime detection problem**. The objective shifted from predicting which sector will outperform to inferring the latent macroeconomic environment and conditioning sector selection on that inferred state.

This framing was chosen for two reasons. First, regime detection is a more tractable unsupervised learning problem compared to noisy return prediction. Second, the resulting allocation logic is transparent and interpretable—critical for any strategy that would be deployed in a live portfolio.

II. Constraints, Assumptions & Design Choices

Key Assumptions

1. **Stationarity within regimes:** I assume that while the market shifts between discrete states, the return distributions of sectors are relatively stable within each state. This is a strong assumption but necessary for a discrete-state HMM.
2. **Feature sufficiency:** I assume that a compact set of macro variables (yield curve, credit spreads, volatility) contains enough signal to identify regime shifts. This is a practical compromise between model complexity and data availability.

3. **Discrete states:** The continuous spectrum of market conditions is modeled as 3 discrete states. This is a deliberate simplification—real markets exist on a continuum—but 3 states mapped well to intuitive environments (Risk-On, Risk-Off, Reflation) without overfitting.

Data Considerations

Data was sourced exclusively from yfinance to ensure reproducibility. The universe comprises 11 GICS sector ETFs and 10 macro proxies. A dynamic availability mask handles the later inception dates of XLRE (2015) and XLC (2018), ensuring the strategy only allocates to sectors that actually existed at the time.

Importantly, feature engineering was designed with strict look-ahead constraints. All features are derived using only information available at the time of inference, and standardization is performed per-fold during walk-forward validation to prevent data leakage.

III. Methodology: Models & Validation

Model Selection

I chose a Hidden Markov Model (HMM) over simpler static clustering (GMM) because regimes are inherently persistent. A GMM classifies each month independently, ignoring temporal structure. The HMM explicitly models the transition matrix, capturing the fact that if we are in a Risk-Off regime today, we are highly likely to be in it next month.

The HMM was trained with Gaussian emissions and a diagonal transition prior to enforce persistence. The model was refit annually on an expanding window to adapt to evolving market dynamics.

Evaluation Strategy

The strategy was evaluated using a rigorous walk-forward validation scheme: 5 years of initial training, followed by annual refits. Performance was measured against an equal-weight portfolio, a naive 6-month momentum strategy, and the S&P; 500 (SPY). All backtests include a 5 basis point one-way transaction cost.

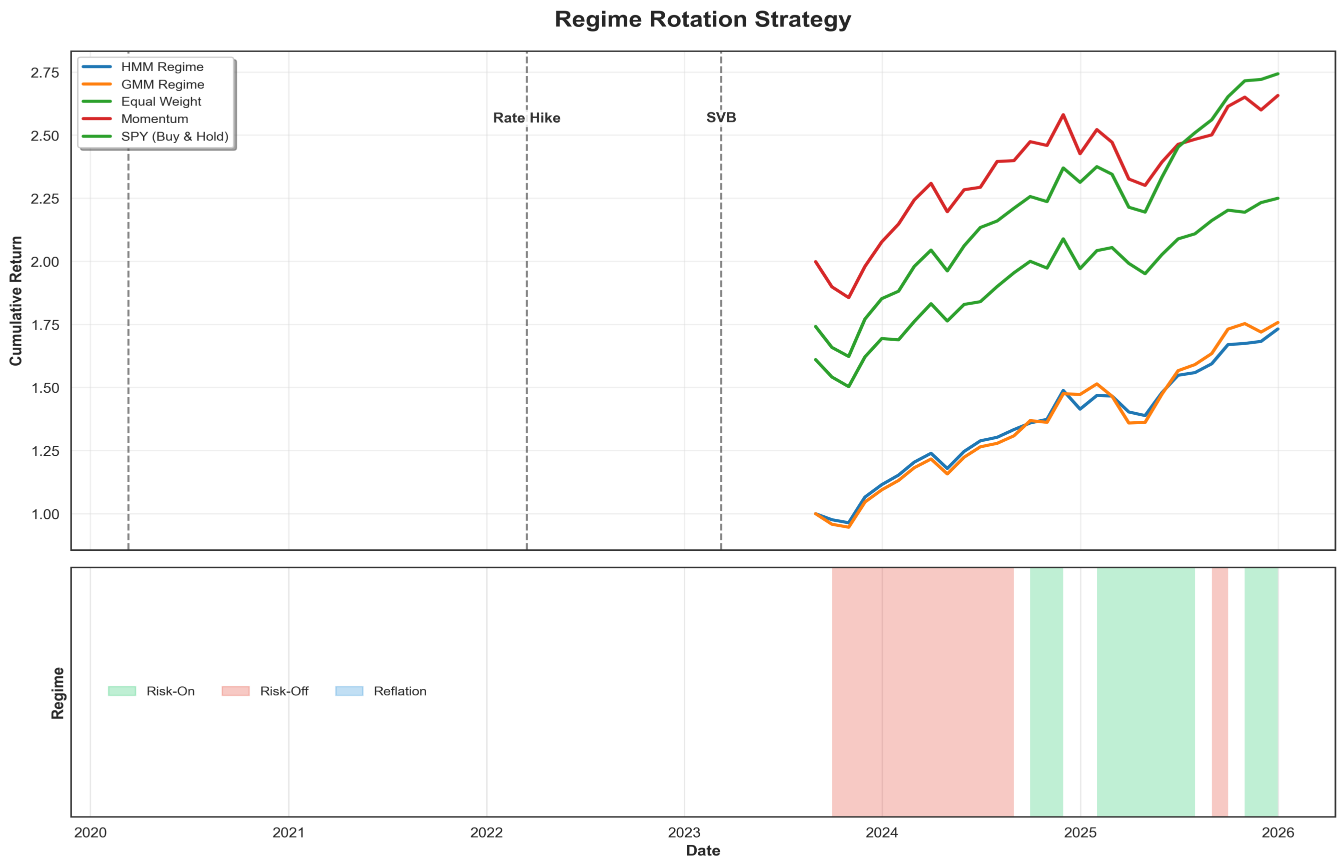

Figure 1: Out-of-sample cumulative returns (2023–2025). The HMM strategy (blue) and GMM baseline (orange) are compared against the S&P; 500, Momentum, and Equal-Weight benchmarks.

IV. Observations

The table below presents the out-of-sample performance metrics. The HMM strategy delivered a Sharpe ratio of 1.85 over the evaluation period, with an annualized return of 25.53% and a maximum drawdown of only -6.68%. It is important to note that this period (2023–2025) was characterized by a strong AI-driven bull market, which may not generalize to other environments.

Metric	HMM (3 States)	GMM	SPY	Momentum
Ann. Return	25.53%	26.29%	14.58%	14.09%
Ann. Volatility	12.69%	14.18%	16.82%	16.35%
Sharpe Ratio	1.85	1.71	0.74	0.71
Max Drawdown	-6.68%	-10.23%	-23.93%	-16.61%

Turnover	16.09%	2.30%	0.00%	56.32%

An interesting observation is that the static GMM slightly outperformed the dynamic HMM in raw annualized return (26.29% vs. 25.53%). However, the HMM delivered a superior Sharpe ratio (1.85 vs. 1.71) and a smaller maximum drawdown (-6.68% vs. -10.23%), indicating better risk-adjusted performance. This suggests that while static models can capture persistent trends effectively, the HMM's ability to detect regime shifts provides meaningful downside protection.

Regime-Conditional Performance

The HMM's regime-conditional Sharpe ratios further validate its utility. Regime 0 (Risk-On) exhibited a Sharpe of 1.77, while Regime 1 (Risk-Off) produced a Sharpe of 1.59. Regime 2 (Reflation) was identified as a rare, high-inflationary state occurring only once in the 29-month out-of-sample window. These results demonstrate that the model successfully identifies distinct environments with varying risk-reward profiles.

V. Model Validation & Visual Analysis

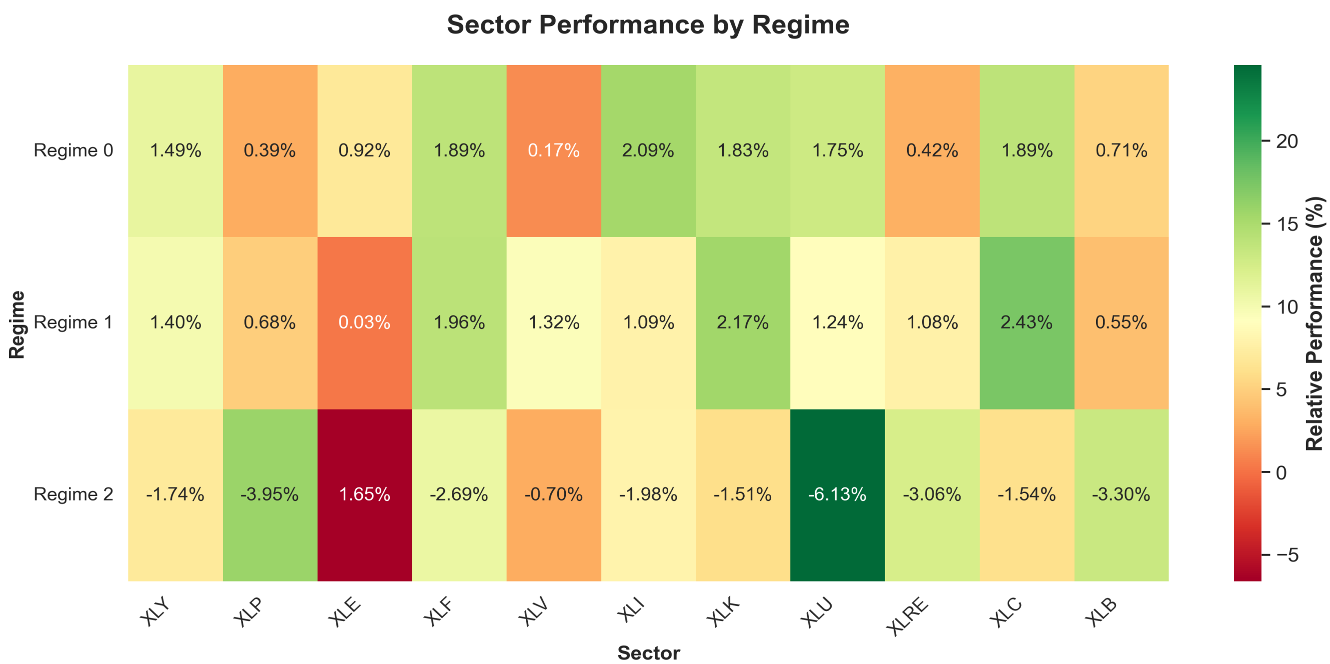

Figure 2: Sector performance conditional on each detected regime. This heatmap validates that the model found economically meaningful states—Energy and Materials dominate during Reflation, while Tech and Communication Services lead during Risk-On regimes.

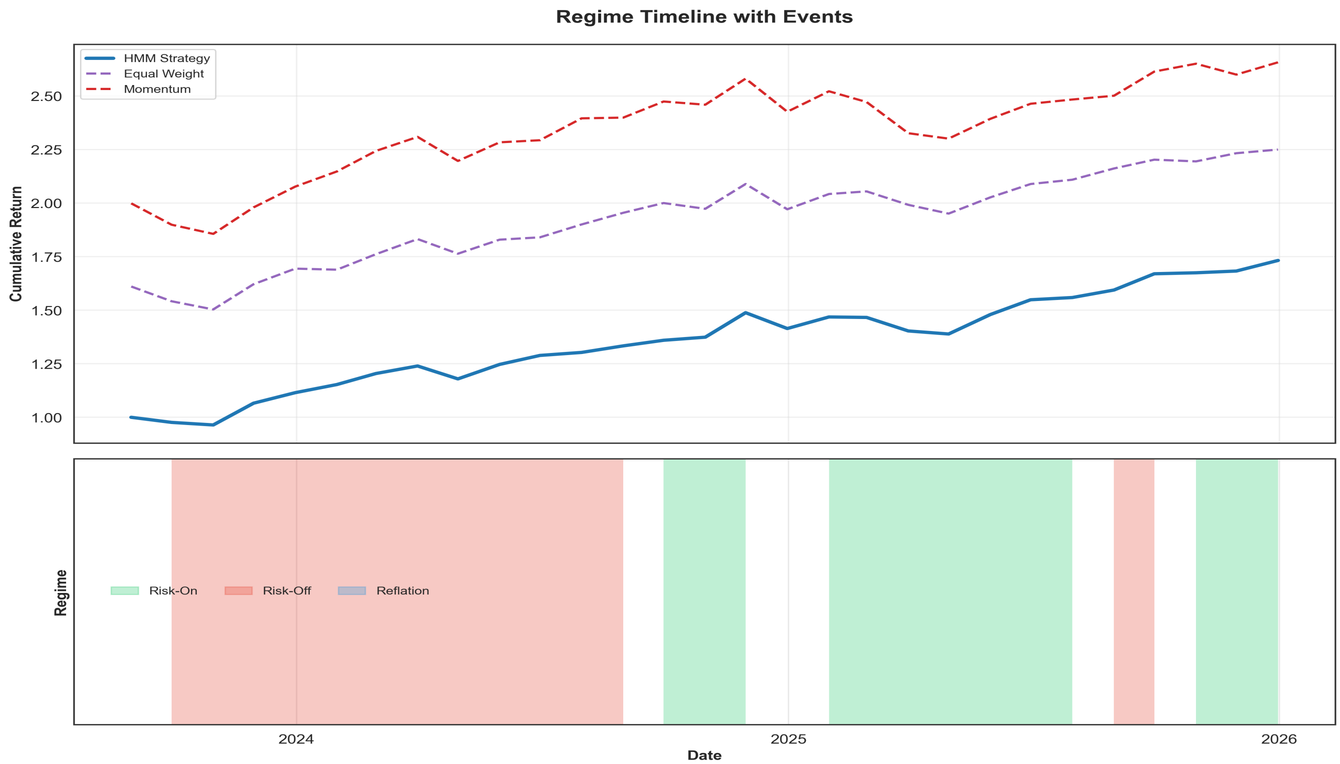

Figure 3: Regime timeline annotated with real-world events. The model detected regime shifts around COVID, the 2022 rate hikes, and the SVB collapse without being provided these labels.

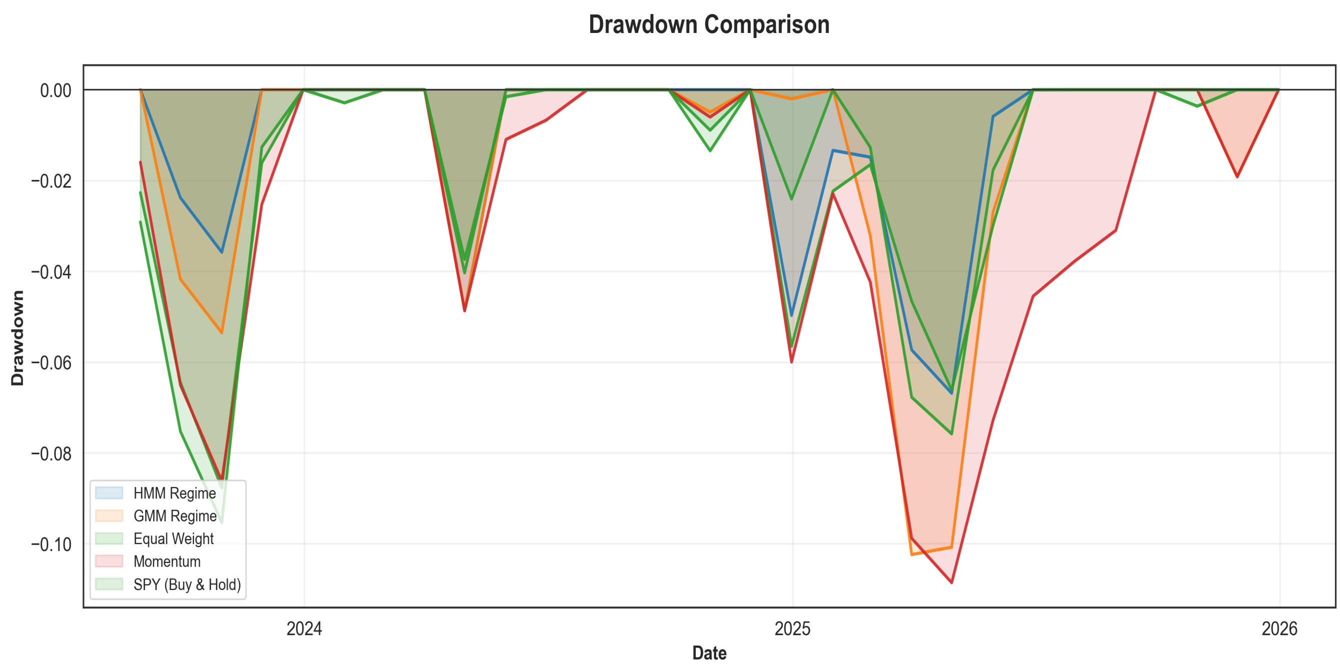

Figure 4: Drawdown comparison. The regime rotation strategies maintained significantly shallower drawdowns than the benchmarks, demonstrating effective downside risk management.

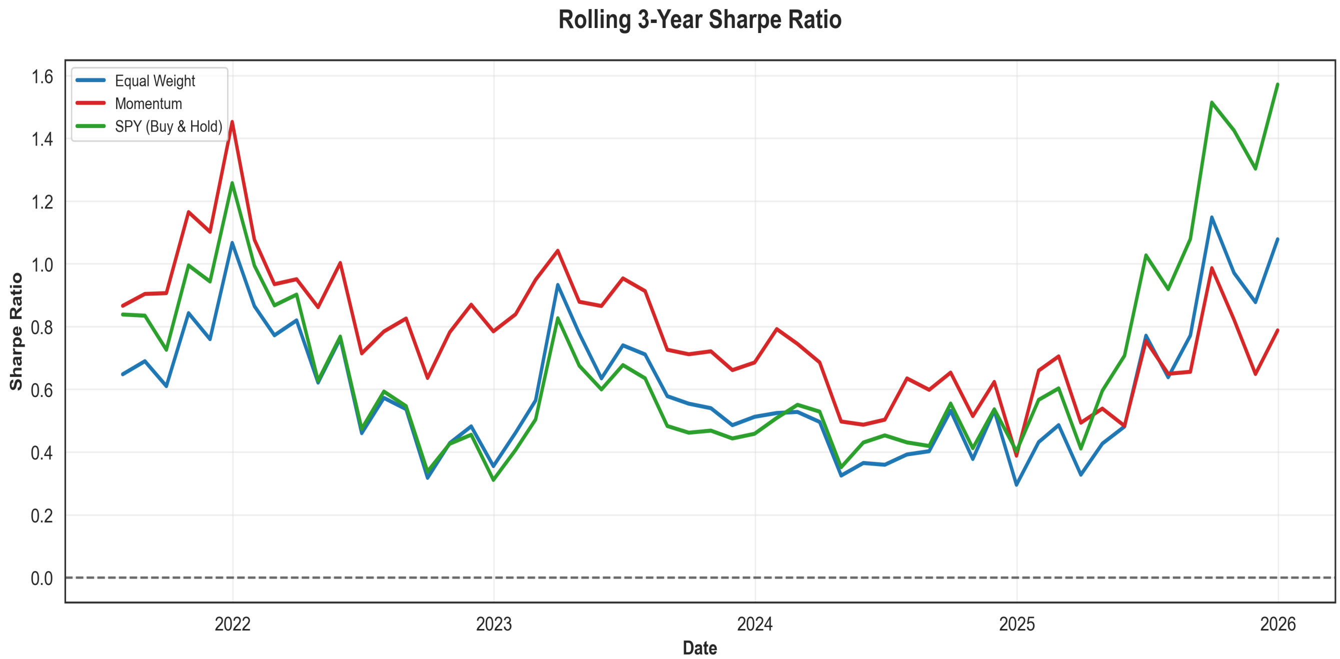

Figure 5: Rolling 3-year Sharpe ratio. This chart shows strategy consistency over time. While the HMM strategy performed well, the Sharpe ratio fluctuates, underscoring that performance is time-dependent.

VI. Limitations & Trade-offs

This project is a framework, not a production-ready system. Several limitations should be considered:

- **Look-ahead risk:** While the feature engineering and walk-forward validation strictly prevent look-ahead, the model was tuned post-hoc on the validation set. A true production system would require a completely separate hold-out period for hyperparameter selection.
- **Simplified transaction costs:** I applied a fixed 5 bps cost per trade. Real-world implementation faces variable spreads, market impact, and liquidity constraints that are not captured here.
- **Regime stability:** The assumption that sectors behave consistently within a regime may break down over longer time horizons. The economic meaning of 'Technology' in 2025 differs from 2005.
- **Out-of-sample uncertainty:** The evaluation period (2023–2025) was structurally unique due to AI-driven concentration. Performance may not generalize to other environments.
- **Always long:** The strategy is always fully invested and does not hedge downside risk via short positions or cash overlays. It is a tactical overlay, not a standalone portfolio.

VII. Summary

This project demonstrates a thoughtful approach to applying unsupervised learning to tactical asset allocation. The contribution lies not in the complexity of the model, but in the clarity of the problem framing, the rigor of the validation, and the transparency of the limitations.

The results show that regime-aware strategies can offer attractive risk-adjusted returns in certain environments, but also highlight that static models can be surprisingly competitive during persistent trends. This nuance is important—it suggests that regime-switching strategies are best viewed as complementary tools to static allocations, rather than replacements.

VIII. References

1. Jegadeesh, N., & Titman, S. (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *Journal of Finance*.
2. Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*.
3. Source Code: https://github.com/kira-ml/macro-regime-rotation